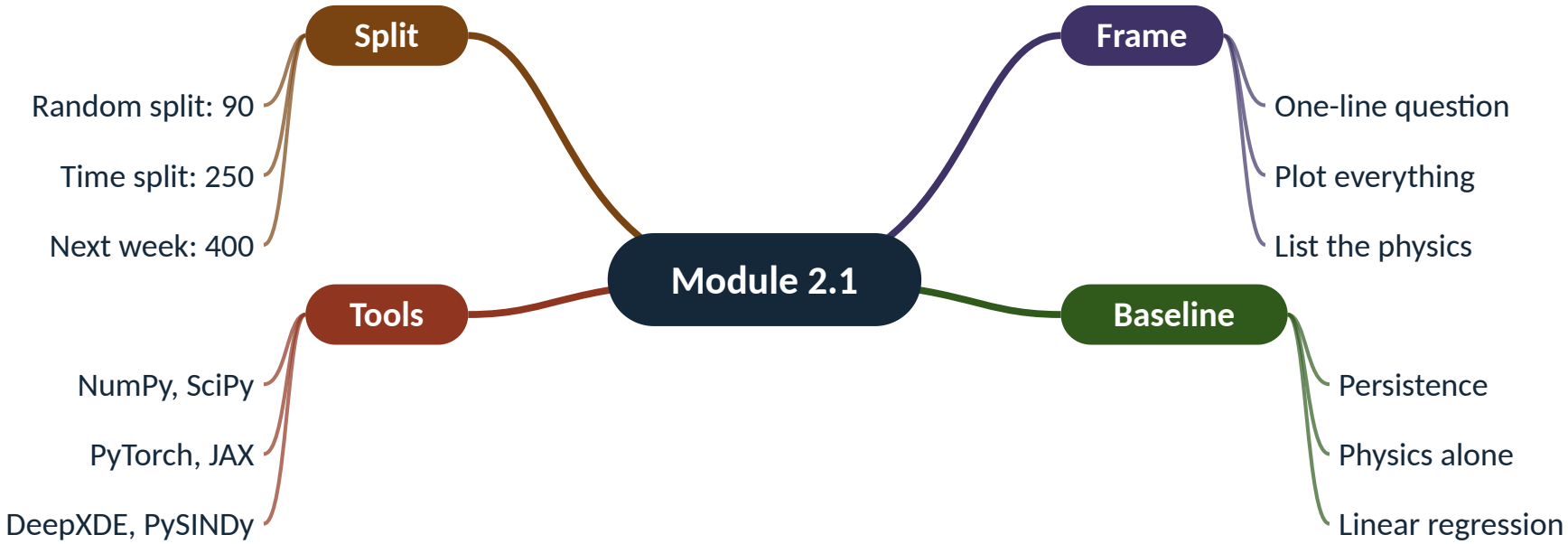
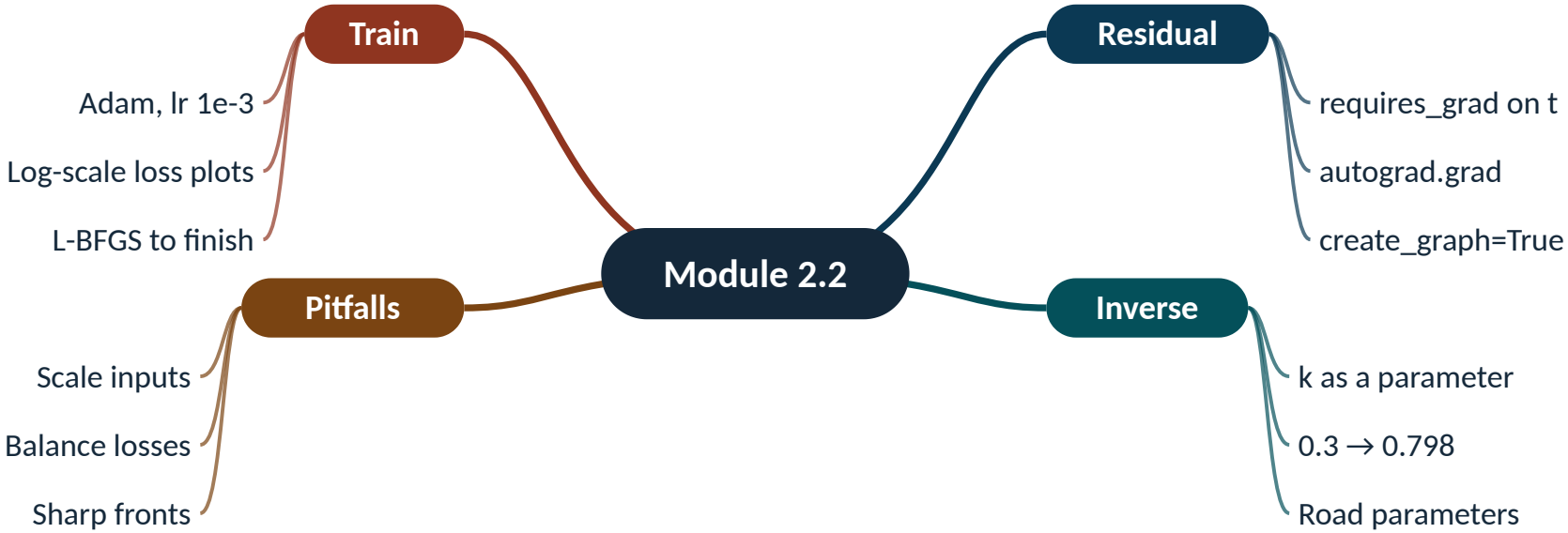
Go 2 mind map


Go 2
2.1 The SciML workflow and toolchain
Frame
Split
Baseline
Tools
2.2 Build a physics-informed neural network, step by step
Residual
Train
Inverse
Pitfalls
2.3 Solving PDEs: classical solvers and PINNs
Finite differences
Stability
Finite volumes
PINN for PDEs
2.4 Discovering equations from data: SINDy and neural ODEs
Idea
STLSQ
Noise
Neural ODEs
2.5 Operator learning: DeepONet and neural operators
Operators
Data needs
Architectures
Uses
2.6 Gaussian processes and uncertainty
GP basics
Calibration
Tuning
Uses
2.7 Hybrid modelling: physics plus learned corrections
Residual hybrid
Results
Patterns
Choosing
2.8 Validation, uncertainty and deployment
Validation
Conformal
Conditional coverage
Deploy

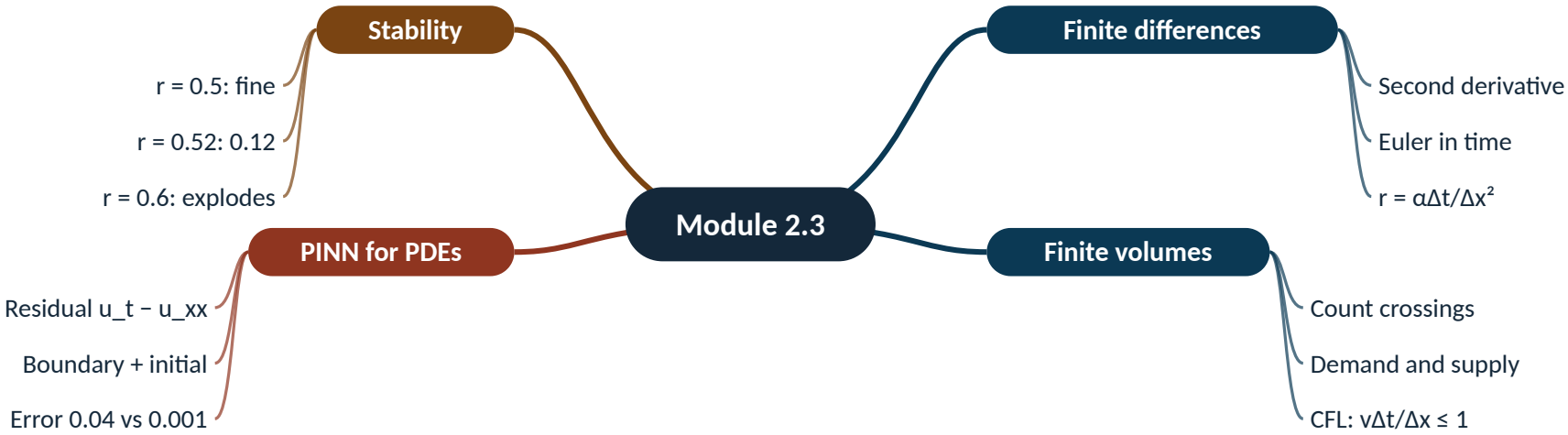
2.1 The SciML workflow and toolchain



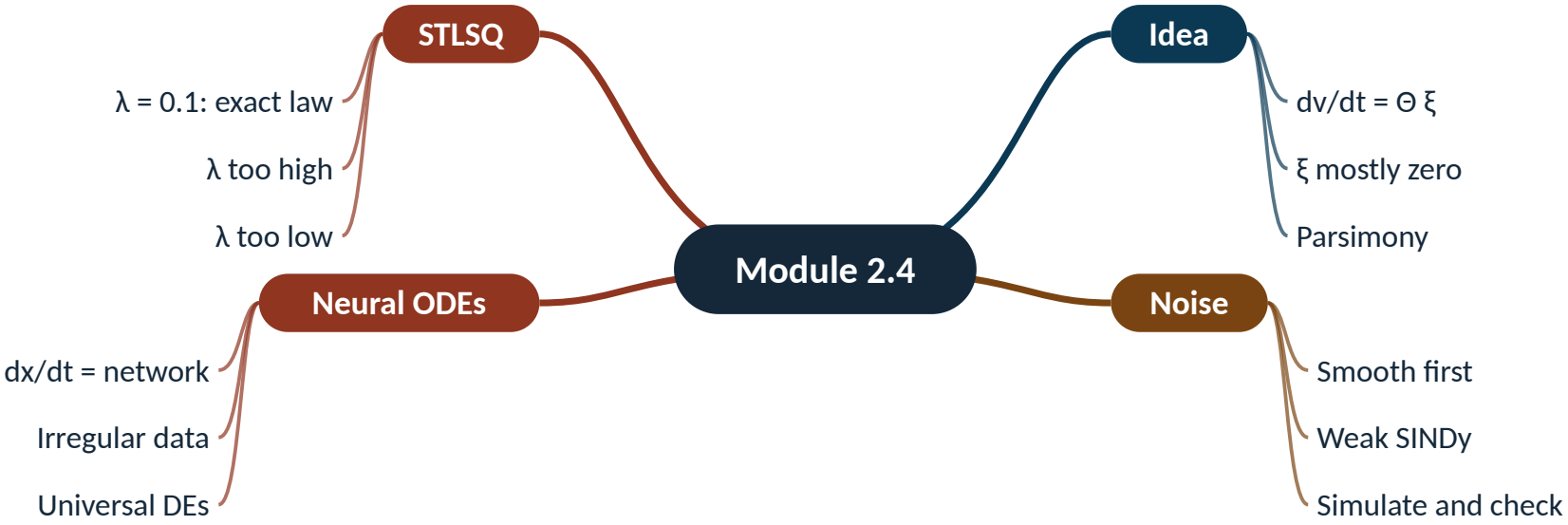
2.2 Build a physics-informed neural network, step by step



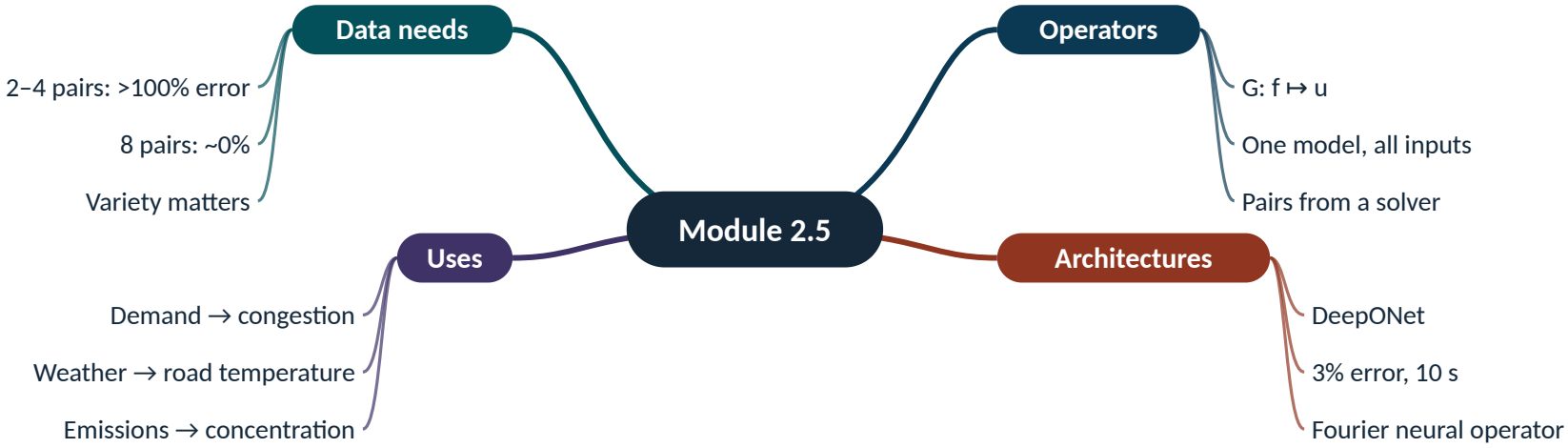
2.3 Solving PDEs: classical solvers and PINNs



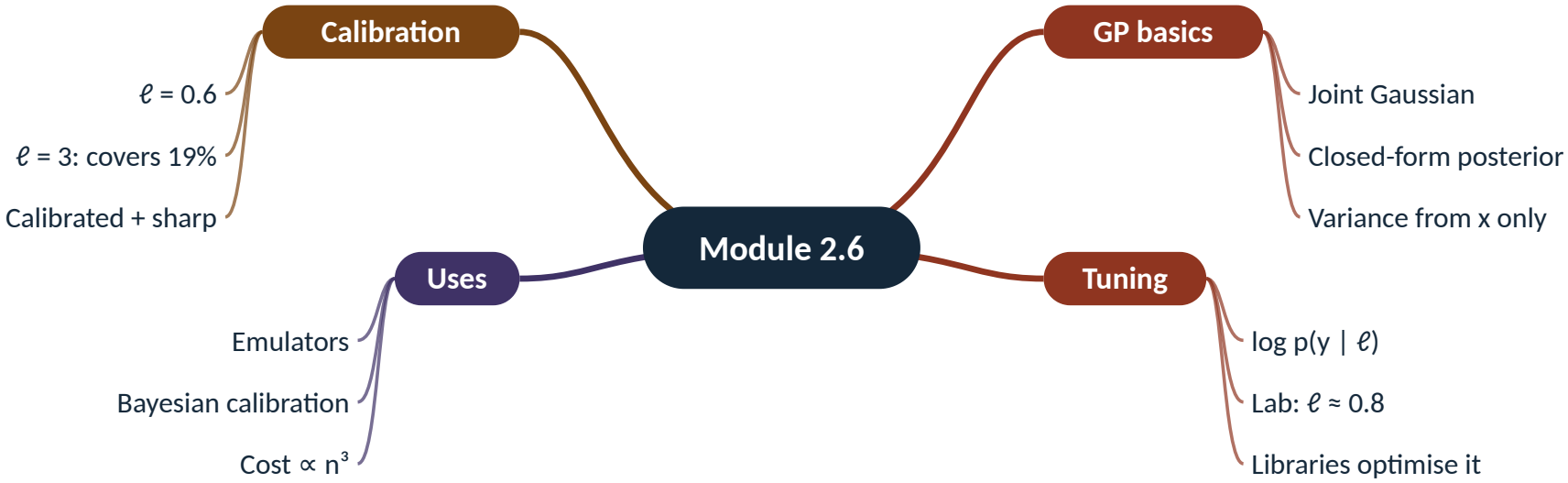
2.4 Discovering equations from data: SINDy and neural ODEs



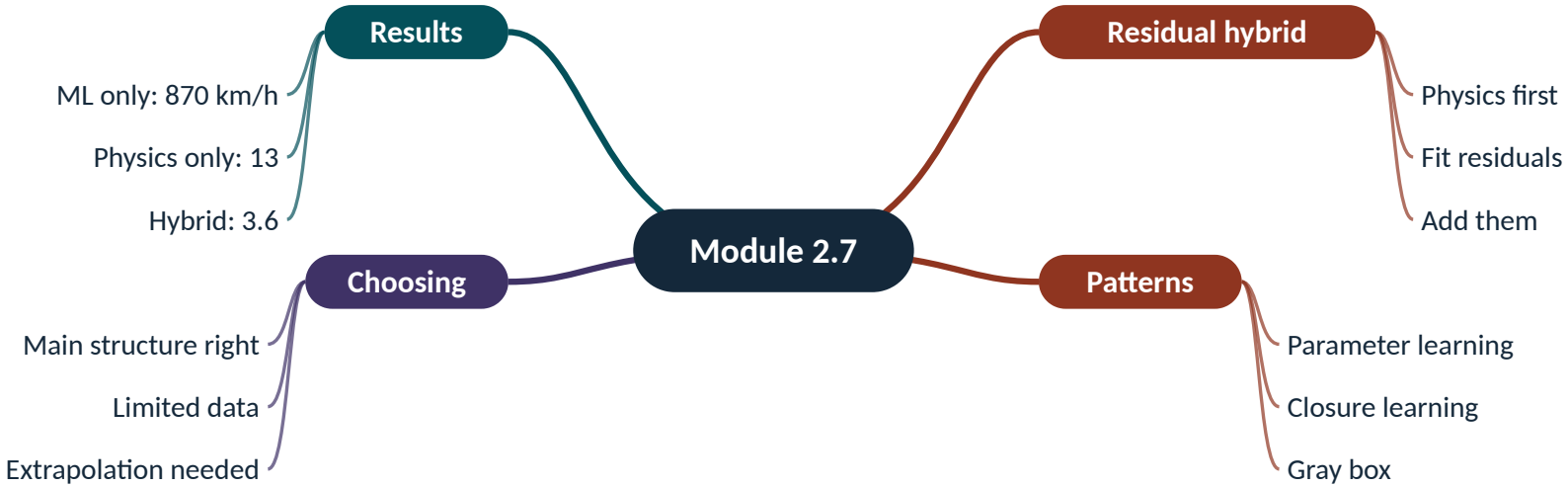
2.5 Operator learning: DeepONet and neural operators



2.6 Gaussian processes and uncertainty



2.7 Hybrid modelling: physics plus learned corrections



2.8 Validation, uncertainty and deployment

